

## UNISIG deep-hole drilling for Military & Defense

Precision gundrilling and BTA drilling for long, straight, tight-tolerance holes in oilfield and aerospace parts. This paper covers what the machine does, the military & defense parts we produce with it, the alloys we run, the tolerances and finishes military & defense buyers expect, and the documentation package.



**Deep Hole** · Long-bore capability — high L/D ratios · gundrilling / BTA deep-hole drilling machine

### Executive summary

**Machine:** UNISIG Deep-Hole Drilling Machine (shop nickname: *Deep Hole*). gundrilling / BTA deep-hole drilling machine at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

**Application:** Precision gundrilling and BTA drilling for long, straight, tight-tolerance holes in oilfield and aerospace parts for Military & Defense customers — Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

**Envelope:** Long-bore capability — high L/D ratios. **Tolerance:**  $\pm 0.0005$ " routine on critical features. **Traceability:** Material by heat and lot on every job.

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### The machine, in context

Deep, straight, concentric holes are one of the hardest features to produce on any CNC platform without a purpose-built machine. Most job shops don't have one. Downhole tool bodies, valve stems with long bores,

hydraulic cylinders, and any part that needs a straight bore longer than a few diameters routinely gets outsourced — or worse, made poorly on an inadequate setup. UNISIG deep-hole drilling is a genuine capability moat.

Full spec table

|                           |                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------|
| Machine type              | UNISIG deep-hole drilling center — gundrill + BTA capable                                                |
| Manufacturer              | UNISIG (Menomonee Falls, WI) — US-built purpose machines                                                 |
| Platform depth capability | From a few inches to over 65 ft (20 m) per drilled hole                                                  |
| Platform diameter range   | Under 1 mm gundrill through ~1 m BTA-class bores                                                         |
| Common configurations     | USC-M series combines gundrilling + BTA + up to 120 tools in one guarded machine                         |
| Reference model USC-3M    | 71" drill depth per side; angled holes at +30° / -15°; 63" × 78" rotating table; workpieces to 66,150 lb |
| Straightness control      | Sub-thousandth-per-foot drift on typical setups                                                          |
| Concentricity             | Held tighter than conventional CNC-lathe drilling can offer                                              |
| Coolant strategy          | High-pressure through-tool                                                                               |
| Materials                 | Inconel, Super Duplex, 17-4 PH, 4140/4340, Monel, titanium                                               |

Who this serves in Military & Defense

**Typical buyer:** Defense primes, subassembly manufacturers, tactical vehicle component OEMs, munitions component suppliers, weapon-system integrators, sustainment (out-of-production replacement) contractors.

**What's hard about military & defense work:** ITAR compliance, security-sensitive prints, program-specific quality plans, tight-tolerance structural components, exotic-alloy selection driven by service environment, and legacy-part sustainment where the OEM no longer supports the design.

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## Typical Military & Defense parts we produce on Deep Hole

- Gun barrel and weapon-component bores
- Hydraulic cylinders for defense platforms
- Long structural components with internal passages
- Sensor tube and probe components
- Actuator cylinder bores

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

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## Alloys we run for Military & Defense

**Standard range:** 17-4 PH · 15-5 PH · 13-8 Mo · 4140 pre-hard · 4340 · Titanium Grade 5 · Inconel 718 · Aluminum 7075 · High-strength steels · Customer-specific alloys on request.

Defense parts routinely spec 4140 pre-hard, 4340, and precipitation-hardening stainless — condition-specific machining knowledge matters, same as aerospace. Titanium is common on weight-critical structural components. Legacy sustainment often specifies materials nobody sees anywhere else; we work with the spec sheet you provide and validate machining strategy before we cut.

**What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

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## Tolerance and finish expectations in Military & Defense

±0.0005" routine on critical features. Tighter with proper setup and fixture. GD&T-driven tolerancing supported. CMM verification on first articles; customer-plan-driven measurement on production runs. Documentation the level a program office expects.

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## Documentation and quality workflow

ITAR-controlled work under NDA with customer-specific document control — prints stay in-shop. Customer-specific quality plans, PPAP, and first-article documentation standard. Material traceability by heat and lot on every job. Certificates of conformance with delivery.

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## Response time and emergency work

Program-driven schedules honored. Emergency component replacement supported for sustainment programs with established relationships. Legacy-part reverse-engineering supported for out-of-production sustainment.

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## How we compare

Compared to a large defense-prime captive shop: we're faster for one-offs, small runs, and legacy sustainment where the prime's supply chain has moved on. Compared to a lowest-bidder commodity shop: we handle ITAR, we document properly, and we understand that a program-office audit is a real thing.

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## Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Military & Defense work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

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